

AI Assistant

BTFViewer's AI Assistant helps you investigate RTOS traces.

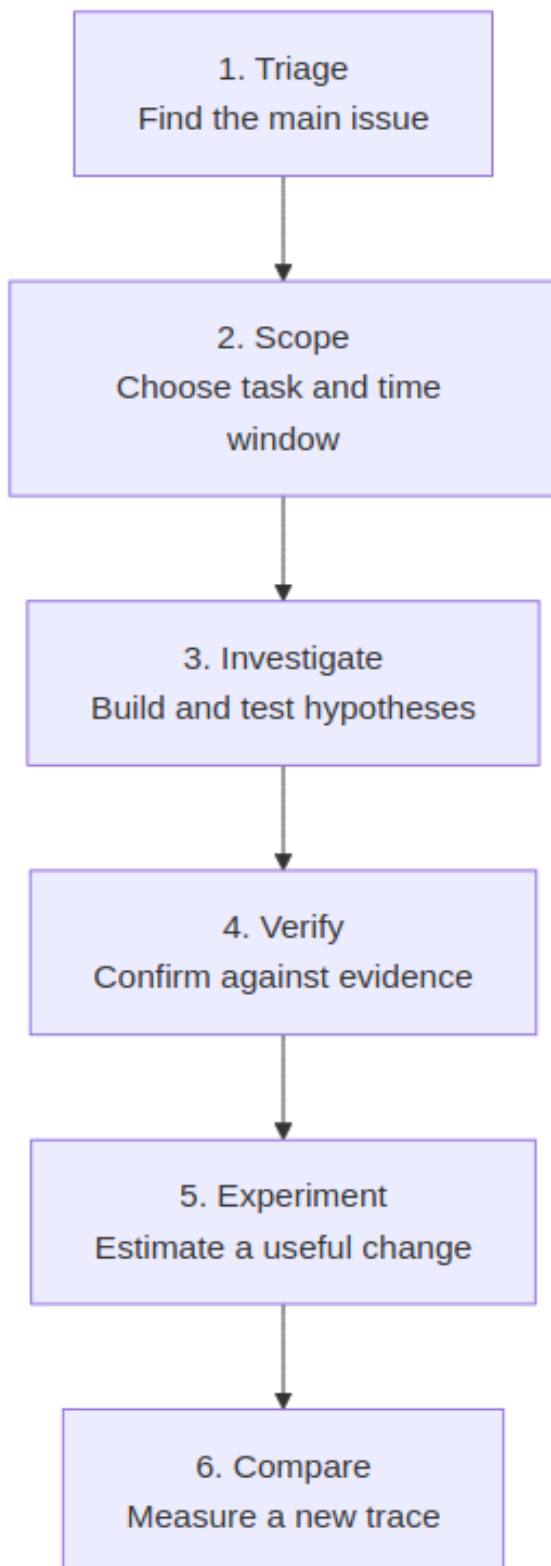
It organizes measured evidence, tests possible explanations, and guides you back to the relevant timeline region.

Scope: AI works with BTFViewer Findings, Statistics, timeline queries, and Trace Compare results. It does not read firmware source or ELF files. `what_if` results are heuristic estimates. They are not RTOS scheduler simulations or measured trace data.

Where to start

Your goal	Read
Learn the product and open the AI panel	README.md → AI Assistant
Follow a repeatable diagnosis procedure	WORKFLOWS.md
Understand a metric or Statistics page	STATISTICS.md
Configure, evaluate, or implement the AI system	This document

For a first investigation, use this sequence:



The key rule is simple: **do not jump from a finding directly to a mitigation.** Scope the incident, investigate the cause, verify the evidence, then experiment and compare.

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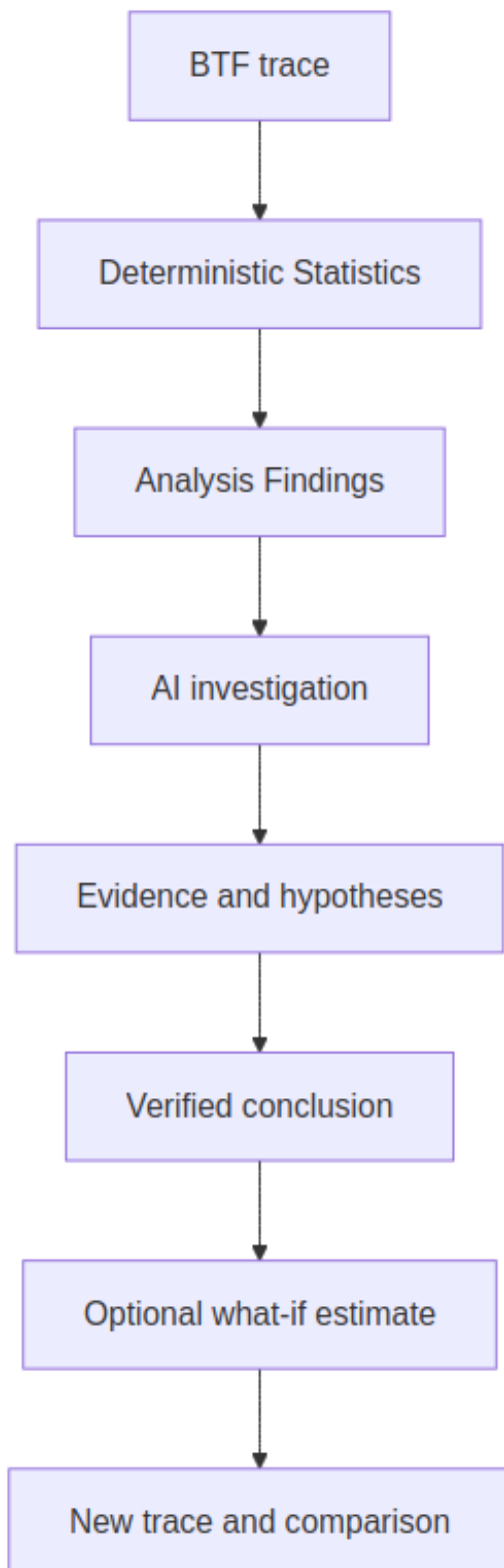
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Overview

This section explains what the AI Assistant does, what evidence it uses, and where its responsibility ends.

Data flow and responsibility



The AI receives structured Findings and summary metrics. It does not receive the complete raw event stream.

When more detail is needed, it requests scoped evidence through the **GUI tools**. This evidence can include:

- per-task metrics;
- timeline search results;
- correlations;
- critical paths;
- Trace Compare tables.

The AI still does not read the raw `.btf` file directly.

AI can explain evidence, find correlations, rank possible causes, challenge assumptions, and make estimates.

Deterministic Statistics and the timeline remain the source of truth.

What the panel does

- **Start Investigation** runs **Auto investigate** when the log is empty.
- The stepper tracks **Triage** → **Scope** → **Investigate** → **Verify** → **Experiment** → **Compare**. Select a completed stage to return to its output.
- **Investigate**, **Root cause**, **Verify finding**, **Auto investigate**, **What-if**, **Optimize**, and **Diagnostic report** display an Investigation plan.
- **Clear** removes the conversation, resets usage, and clears the current investigation.
- The usage bar shows **Context: Compact · 4.6k tok · 3 tools · 12s** (mode, tokens, tools, and model time). **Settings** → **AI** → **Context** chooses Compact, Balanced (default), or Full evidence.
- A non-empty `investigation_session` restores after restart only when the log still has a user or assistant turn. An empty or cleared log does not restore a Current Issue card.
- Read-only tools and `export_report / export_investigation` run immediately. GUI-changing actions wait for **Apply** unless **Auto-apply GUI actions** is enabled. Desktop exports open a save dialog; the web app downloads the file.

Toolbar **Compare** becomes available when at least two traces are open. **Query with AI...** sends the Trace Compare tables rather than the current Findings. **Save as baseline** and **Score vs baseline** use the same stored profile as `baseline_score`.

Ctrl+K provides quick access to Analysis, AI, Compare, workspace presets, and Inspect task.

Scoping an event or region

Entry point	Scope
Timeline segment → Ask AI about this event	The selected task, core, and segment around <code>jump:TIME</code>
Timeline → Explain this region with AI	Available with at least two cursors; uses C1-Cn
AI panel → Explain region	Uses C1-Cn when available; otherwise uses full-trace Findings

Enable **Limit to C1-Cn** when diagnosing a phase-specific issue. The prompt then includes `Cursor region window: jump:lo ... jump:hi`, and every cited `jump:TIME` should remain inside that interval.

Getting started

Start here if you are using the AI Assistant for the first time.

This section explains how the AI supports common investigations. For symptom-to-metric playbooks and exact ask order, use [WORKFLOWS.md](#).

First investigation

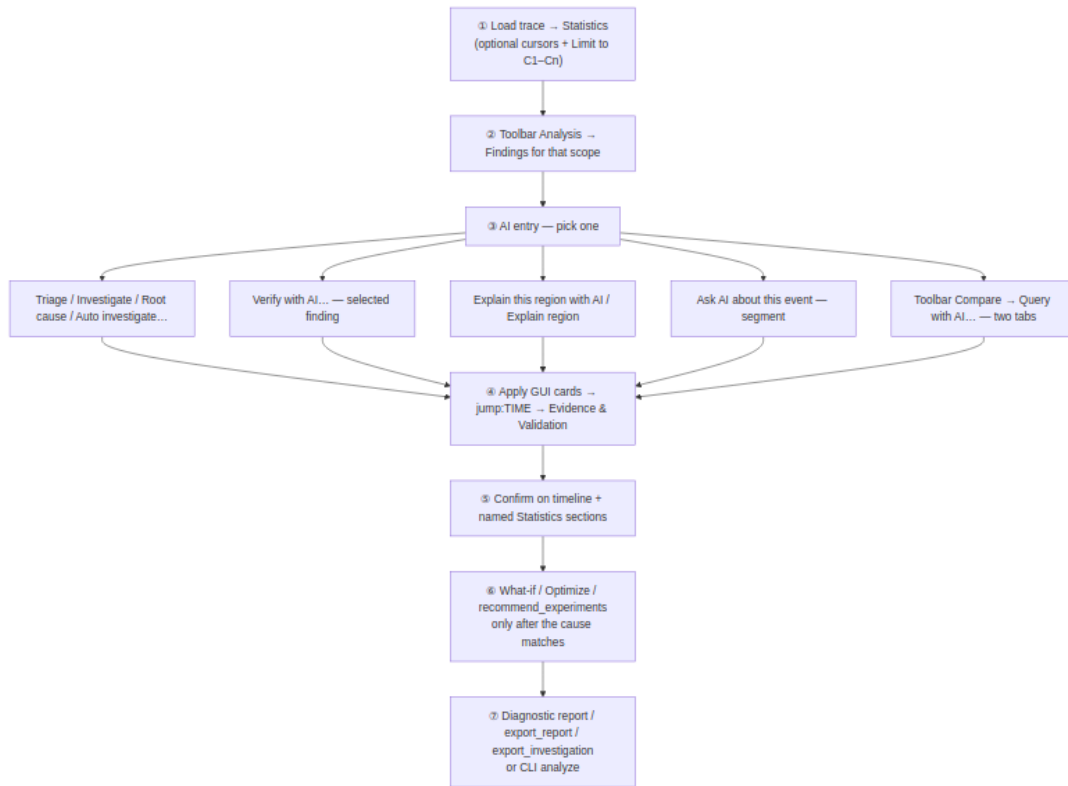
Do not start by choosing individual tools.

Start with the main user actions. Let **Investigate** select deeper evidence tools when they are needed.

Step	Action	Expected result	Check before continuing
1. Triage	Triage findings or toolbar Analysis	Ranked Critical, Warning, and Info issues	The named Statistics page shows the same issue
2. Scope	Select a finding and place or apply C1-Cn	One task, incident, or time window	Enable Limit to C1-Cn for phase-specific questions
3. Investigate	Investigate ; use Root cause when a suspect is known	Hypotheses, correlations, dependencies, and critical paths	Open the cited jump:TIME, range:L0/HI, and Statistics pages
4. Verify	Verify with AI... or continue the plan	Supported, rejected, or insufficient verdict	Confirm scope, task names, times, contradictions, and alternatives
5. Experiment	What-if, Optimize , or an experiment plan	Ranked estimated changes	Treat results as estimates; change the system and capture a new trace
6. Compare	Open before/after traces → Compare	Measured deltas and experiment verdict	Use equivalent workloads and comparable cursor scopes

For normal use, remember only these actions: **Triage findings**, **Investigate**, **Verify with AI...**, **Explain region**, **What-if / Optimize**, and **Trace Compare**. Function names such as `correlate_events` and `rank_root_causes` are advanced and implementation references.

End-to-end flow



Do not ask for a fix before the timeline supports the finding.

Empty Statistics or the wrong scope can produce an answer that sounds confident but has weak evidence.

Prefer built-in templates. They already use the expected metrics and units.

Investigation workflows

Use these workflows after you have selected the task, finding, or time window to investigate.

Investigation workflow

Step	Template or tool	Why
1	Triage findings / detect_anomalies	Rank Critical / Warning / Info; open Timeline Anomalies / Worst Events / Task Health
2	Investigate / investigate — or Findings Auto investigate...	Root-cause chain, hypotheses, alternatives, suggested tools
3	Verify finding / Findings Verify with AI...	Confirmed / Rejected / Inconclusive with jump:TIME evidence
4	correlate_events + query_raw_metric	Merge blocking / execution / migrations / sync / PI for one task
5	find_critical_path / detect_priority_inversion	Preempt/block path; L/M/H inversion suspects
6	build_task_dependency_graph / analyze_temporal_causality	BTF wait/preempt/migrate chain
7	rank_root_causes / challenge_conclusion	Rank then alternatives before what_if
8	find_related_findings / compare_tasks	Adjacent findings; side-by-side task deltas
9	set_cursors / zoom_to_range / highlight_task / bookmark_finding	Narrow the timeline (Apply cursors unless auto-apply is on); click range:L0/HI / btfrange: on critical-path evidence
10	Evidence & Validation panel	Status, direct-evidence table, checks, missing evidence, next action; details hold quality/cost/trees
11	investigation_replay / generate_report / export_investigation	Structured close-out; optional export_report

Root cause walks deadline/WCET → preemption → blocking → mutex → inheritance → migration for the top finding. Use it when triage already named a suspect task.

Auto investigate chains verify-style steps for one finding (investigate → correlate

→ critical path / graph / temporal → rank → challenge → what-if) and advances the Investigation plan checklist. Use **Verify** when you already have a finding id and want a short verdict.

Explain region and Ask event

Entry	When it appears	Scope
Timeline → Explain this region with AI	Only with ≥ 2 cursors ; grayed when AI is off	C1-Cn; prompt gets Cursor region window: jump:lo ... jump:hi
AI panel → Explain region	Always	Same window when ≥ 2 cursors; full-trace Findings if none
Timeline segment → Ask AI about this event	Segment under the pointer; grayed when AI is off	One task / core / segment around jump:TIME

Stay inside the stated window: every jump:TIME in the reply should fall between C1 and Cn (or the model should say the window has no matching evidence). Enable **Limit to C1-Cn** so Statistics / Findings / query_raw_metric match that window. Clicking a jump:TIME outside the cursors usually means the model invented or reused a full-trace time — discard it and re-ask with cursors + scoped Findings.

What-if and optimize workflow

what_if and optimize_experiment are **heuristic slice-replay** tools: they reallocate measured execution slices, scale migrations / blocking, and adjust core-util balance. They do **not** simulate an RTOS kernel or a deterministic scheduler. Every result carries a disclaimer. After a promising estimate, recommend_experiments suggests validation steps (simulation / firmware / measurement).

Goal	What to run	Typical change phrases
One concrete idea	What-if → what_if	pin CS[28] to Core_0, raise priority of Low[266], reduce mutex contention 50%
Rank several ideas	Optimize → optimize_experiment (then optional optimize for qualitative notes)	Host picks pin-to-dominant / quiet core, contention –50%, priority up, migrations –50%
Soft advice only	optimize	Finding-text mitigations without scored experiments
What to try next on the bench	recommend_experiments	Validation experiments from findings heuristics

How to read the result:

Compare baseline with simulated. Check:

- migrations;
- blocking_ns;
- load_balance_score;
- the deltas.cost ranking.

A lower cost is better in the Experiment List.

Medium confidence means the change is worth testing on real hardware. **Low confidence** usually means the change description was vague or there were too few slices.

Common use cases

These examples show how to apply the workflow to common RTOS trace problems.

Use cases

Situation	Before you ask	Template / tools	Then verify
Unknown — first look	Full-trace or cursor-scoped Findings	Triage findings → Investigate	Timeline Anomalies / Worst Events / Task Health; jump:TIME
Hottest / noisiest task	Findings name a suspect	Task profile	Period / Jitter; Task Health; Task × Core; Execution / Blocking p95/p99
Tick jitter / missed ticks	Trace Health in scope	Tick health	Tick Distribution (not Period / Jitter — that page is task inter-arrival)
Confirm one finding	Select it in Analysis Findings	Verify with AI... / Verify finding	Evidence panel; timeline
Explain a time window	≥2 cursors; Limit to C1-Cn on	Context menu or Explain region	Only jump:TIME inside C1-Cn
One segment / ISR slice	Right-click the segment	Ask AI about this event	That task's row; nearby STI
Auto walk a finding	Select finding → Auto investigate...	auto_investigate	Investigation plan + Evidence

Situation	Before you ask	Template / tools	Then verify
Migration thrash / ping-pong	Scope thrash window; Findings mention the task	Migration thrash → correlate_events → What-if pin / Optimize	Task × Core; Timeline Anomalies migration bursts; Migrations Rate/Ping; Heatmap / Chord; Core Affinity
Priority inversion / PI boost	Inheritance finding or PI episodes in scope	Priority inversion / detect_priority_inversion → find_critical_path	Priority Inheritance; Mutex hold; Waiter × Owner (heuristic handoff)
High blocking / mutex wait	Scope the stall; suspect task known	Highest latency → query_raw_metric blocking/sync → What-if contention / priority	Worst Events; Waiter × Owner; Blocking p95/p99; Mutex hold; Priority Inheritance
WCET / deadline pressure	Thresholds set in Display → Analysis	WCET / hot CPU or Deadline / budget → check_budget	Timeline Anomalies / Worst Events; Period / Jitter; Task Health; Execution Max / p95 / p99; Deadlines

Situation	Before you ask	Template / tools	Then verify
Compare two tasks	Both names known	compare_tasks	Execution / Blocking / Migrations side-by-side
Related findings	One finding selected	find_related_findings	Shared task / metric / nearby times
Load imbalance across cores	Multi-core util in Findings	Core balance → analyze_traces (multi-tab) or What-if pin to quiet core	Task × Core; Load Balance Score; Concurrent Active; Core Time Breakdown
A vs B build regression	Two tabs open	Trace Compare (toolbar Compare → Query with AI...) / compare_performance / regression_explain	Compare summary strip; Trace Compare pages; same scope on both builds
Drift vs saved baseline	Baseline profile stored (rc / localStorage)	baseline_score	Flags z >2; re-capture if needed
Rank all open traces	≥2 loaded tabs	analyze_traces	Best tab vs Migrations / LB / missed ticks

Situation	Before you ask	Template / tools	Then verify
Write-up for a review	Cause already confirmed	Diagnostic report → generate_report → export_report / export_investigation	Saved HTML/CSV/JSON; evidence times bookmarked
CI gate vs baseline	Desktop CLI	analyze with --fail-on-regression (optional --ai)	Exit code + Markdown narrative

Worked examples

Migration thrash → pin affinity

1. Place cursors on the thrash window; enable **Limit to C1-Cn**; re-open Analysis.
2. Run **Migration thrash** or **Investigate** until the hot task (e.g. CS[22]) and cores match the heatmap.
3. Ask **What-if**: *pin CS[22] to its dominant core* (or run **Optimize** for ranked candidates).
4. Read Δ migrations / Δ load_balance_score. If migrations drop but LB worsens sharply, try pin-to-quietest via optimize_experiment and compare ranks.
5. On firmware: set affinity / reduce bounce; re-capture a .btf and toolbar **Compare** the before/after tabs.

Mutex contention → shorter critical section

1. **Highest latency** / correlate_events for the waiter; confirm hold episodes in Mutex / Priority Inheritance.
2. **What-if**: *reduce mutex contention 50% for TASK* (or another %).
3. Expect lower blocking_ns in the simulated payload — still an estimate. Confirm by shortening the hold in code and re-tracing.

Two builds → regression narrative

1. Open baseline and candidate as tabs; match cursor scope if needed.
2. Toolbar **Compare**, then **Query with AI...** (**Trace Compare** template), or compare_performance then regression_explain.
3. Act only on High/Medium confidence deltas that Statistics on both tabs reproduce.
4. Optional: optimize_experiment on the candidate's hottest task to sketch mitigations (still heuristic).

Cursor window → Explain region

1. Place **C1** / **C2** on the phase of interest (e.g. 1.060 s ... 1.120 s); enable **Limit to C1-Cn**; re-open Analysis.
2. Right-click the timeline → **Explain this region with AI** (or AI panel → **Explain region**).
3. Confirm the user turn lists Cursor region window: jump:lo ... jump:hi. Reject any jump:TIME outside that window.
4. Follow up with correlate_events / query_raw_metric on tasks named in-window; bookmark evidence times.

Priority inversion finding → Verify

1. Open Analysis Findings; select a Priority Inheritance / inversion row.
2. **Verify with AI...** (or **Auto investigate...** for a longer chain).
3. Expect detect_priority_inversion / query_raw_metric (priority_inheritance) / find_critical_path; read the Evidence score and investigation tree.
4. Click in-scope jump:TIME links; confirm L/M/H on the timeline and in Priority Inheritance statistics.

Simulator limits

Does	Does not
Replay measured slices / migrations / blocking gaps for the Statistics scope	Run RTOS scheduling, ISRs, or cache models

Does	Does not
Score pin / priority / contention / migration experiments	Guarantee WCET or deadline after a firmware change
Label every result as estimate / not measured	Replace timeline verification or a new capture

Phrase changes as **pin / affinity / priority / mutex / migration** so the simulator engages; vague text falls back to a qualitative estimate (simulator: none).

Understanding AI results

AI output is an interpretation of measured evidence. Use this section to understand evidence, validation, and confidence before accepting a conclusion.

Evidence and validation

Important conclusions should include:

- evidence links such as jump:TIME, range:L0/HI, and named metrics;
- confidence: **High**, **Medium**, or **Low**;
- evidence quality: **Directly observed**, **Strong correlation**, **Possible explanation**, or **Insufficient evidence**;
- alternative explanations and what would disprove the conclusion.

The Evidence & Validation panel shows:

- conclusion **Status** (Confirmed / Correlated / Suspected / Not observed / Insufficient data);
- **Finding**, a clickable **Direct evidence** table, and **Interpretation**;
- **Checks**, alternative explanations, **Missing evidence**, and one **Next action**;
- **Investigation details** for quality band, cost, tool reasons, and trees.

The Evidence Quality band (under Investigation details) is a diagnostic heuristic. It is **not a probability**.

After the final reply, the host validator checks task names and timestamps. It flags unknown task names and timestamps outside the cursor window.

Prefer built-in templates. They already select the relevant metrics and Statistics pages. Use natural-language questions such as “find STI wait around TaskA” when needed; the host routes them through `search_timeline`. **Analysis Findings** can triage overall findings. **Explain finding** explains the selected Analysis Finding. Other chips: **Explain region, Investigate, Verify finding, Root cause, Trace Compare, Triage findings, Task profile, Diagnostic report, What-if, Optimize, Highest latency, WCET / hot CPU, Migration thrash, Core balance, Tick health, Priority inversion, Deadline / budget, Auto investigate**. Findings also offer **Save recipe...** and **Story...**

Named Statistics pages the templates cite: Timeline Anomalies, Worst Events, Period / Jitter, Unified Jitter, Recurring Patterns, Task Health, Task × Core, Waiter × Owner, Response Time, Critical Path, Preemption Matrix, Mutex Blocking, Core Utilization Over Time.

Configuration, models, and privacy

Connect an endpoint

You can use any OpenAI-compatible endpoint, including Ollama (<http://localhost:11434/v1>).

Chat requests time out after 120 seconds. **Stop** can cancel a request earlier.

Use a context window of at least **8k** when possible. This gives enough room for a full Findings card and one tool round.

If a local model has a smaller context window, use **Settings → AI → Context → Compact**. Compact reduces Findings, tool schemas, tool rows, and chat history.

The shipped Ollama default is `qwen3.5:9b`:

```
ollama pull qwen3.5:9b
```

Larger local models (qwen3.5:27b, qwen3.8:27b, and gemma4:26b) require more memory and usually run more slowly. More parameters do not guarantee a better BTFViewer investigation result. In the recorded suite, qwen3.8:27b reached the same best Overall score as qwen3.5:9b, but its mean latency was about 20 times higher. Older 7B/14B ids such as qwen2.5:7b stay optional. Avoid 3B-class models for investigations: they often skip native tools, return tool JSON as text, or fail multi-step cases.

Configuration examples are available in [examples/ai](#): [ollama.json](#), [gemini.json](#), [openai.json](#), [deepseek.json](#), [grok.json](#), and [presets.json](#).

Importing a preset fills **Settings** → **AI**, including any checkbox flags defined by the file. Review the values before saving. Each preset keeps its own base URL, model, API key, authentication mode, and TLS setting. Unknown preset names are added to the model list.

Field	Meaning
Authentication	none / API key / Sign in, per preset
Model picker	Refresh the served id list, then pick a model
Self-signed TLS	Desktop Allow self-signed TLS skips certificate checks for that preset

API keys use the same precedence on Desktop and Web: Settings → AI first, then OPENAI_API_KEY, then GEMINI_API_KEY, then OLLAMA_API_KEY.

1. Key entered in **Settings** → **AI**
2. OPENAI_API_KEY
3. GEMINI_API_KEY
4. OLLAMA_API_KEY

A local Ollama endpoint normally needs no key. For a custom endpoint, enter its key in the corresponding preset. Live ai-test XML may use `<api-key env="VAR">`. See [README](#) → [API keys](#) for complete examples.

Choose a model

Capability	Small local	Local 9B+	Cloud
Basic Q&A	✓	✓	✓
Tool calling	△	✓	✓
Investigation (investigate / root-cause chain / hypotheses)	△	✓	✓
Complex reasoning (multi-step correlation, alternatives)	△	✓	✓
Large traces (big Findings card / long chat history)	△	△	✓
What-if / optimize (what_if, optimize_experiment)	✓	✓	✓

✓ reliable · △ inconsistent — works sometimes but often skips native tool calls, hallucinates numbers, or truncates on long context; always verify against the timeline before trusting a result.

The recommendations below are based on the 17-case run recorded on 2026-08-19. Scores and latency can change with the endpoint, hardware, model build, and dataset.

If you...	Use
Want the practical local default without an API key	qwen3.5:9b with Balanced . It passed 15/17 cases at 14.5s/case. Full evidence produced its highest Overall score, 88, but passed 14/17 cases at 16.2s/case.
Want a fast cloud response	gemini-3.5-flash-lite with Full evidence . It scored 83, passed 13/17 cases, and averaged 2.6s/case.
Want the best result from the shipped Gemini models	gemini-3.7-flash with Full evidence . It scored 85 and passed 14/17 cases at 25.0s/case.

If you...	Use
Want a second local comparison and can accept high latency	qwen3.8:27b with Balanced . It scored 88 and passed 13/17 cases, but averaged 325.2s/case. It did not provide a consistent quality advantage over qwen3.5:9b.
Can use an optional cloud model outside the shipped suite	gpt-5.6-sol with Compact produced the highest recorded result: Overall 90, 16/17 PASS, and 10.4s/case. It is not included in the shipped benchmark configuration.
Handle confidential traces	Use local Ollama. The raw trace and extracted evidence remain on the local machine.

Small local models may skip native tool calls and emit a fenced `btftool` block instead. The viewer renders the same GUI cards either way, but investigation-heavy templates need a tool-capable model such as qwen3.5:9b to chain calls reliably.

Credential storage

	Desktop	Web
Where keys live	[ai] *_api_key in btft_viewer.rc next to the viewer	Browser localStorage (btft-viewer-settings-v1)
At rest	Encrypted as enc1:... (machine-bound; not portable to another host)	Plaintext in localStorage — treat as convenience only
Sent to the model	Never as a chat field; only as the HTTP Authorization / API header to the configured endpoint	Same
Clear	Settings → AI → clear key, or delete btft_viewer.rc AI keys	Settings → Reset / clear site data

What leaves the machine

Sent to the configured AI endpoint	Not sent	
Analysis Findings: titles, severities, task names, and heuristic text	Raw .btf or .btf.gz bytes	
Metrics and tool results requested by the model	The complete unrequested event stream	
Scoped timeline search and correlation results	API keys in the prompt body	
Trace Compare tables when requested	—	
User question and short conversation history	—	

	Local Ollama	Cloud endpoint
Trace file stays local	✓	✓
Findings / metrics leave the machine	No (loopback)	Yes — to that vendor
Raw BTF uploaded	No	No
API key required	Usually no	Usually yes

Prefer local Ollama for confidential traces. Redact sensitive task names in annotations before using a cloud preset.

Context mode (token usage)

Settings → AI → Context controls how much evidence is sent with each request. Compact is the token-efficient packing mode. It reduces input tokens; Compact also caps the reply at about 300–500 tokens.

	Compact	Balanced (default)	Full evidence
Findings	Top 5 by severity	Top 12	All in scope
Tool schemas	Current stage + search / raw metric	Stage plus neighbours	Complete catalog
Tool results	10 rows; rest summarised	20 rows	40 rows
Chat history	Investigation summary + last 2 turns	Last 6 turns	Last 20 turns
Diagrams	Only if requested	When useful	When useful
What-if	Top 3 candidates	Top 5	Complete

Compact still keeps the cursor region window, real task names, jump:TIME / range:L0/HI, measurements with units, confidence / evidence quality, what-if disclaimers, and at least one alternative or falsification. If Compact omits a relevant finding, ask for a specific finding id or select a larger mode.

More context did not consistently improve the benchmark result. qwen3.5:9b had its best pass count in Balanced, while gpt-5.6-sol had its best score in Compact. qwen3.8:27b and claude-sonnet-5 also scored lower in Full evidence than in Balanced. Treat Balanced as the general starting point, then use **--compare-context** on the intended model and workload. Select Full evidence when the investigation actually needs the additional findings, tool catalog, or history; do not assume it is always more accurate or faster.

Live ai-test defaults to Full evidence. Use **--compare-context** to measure all three modes, or **--context-mode compact** (or balanced) for a single mode. Settings → Context does not apply to the CLI scorer.

AI tools reference

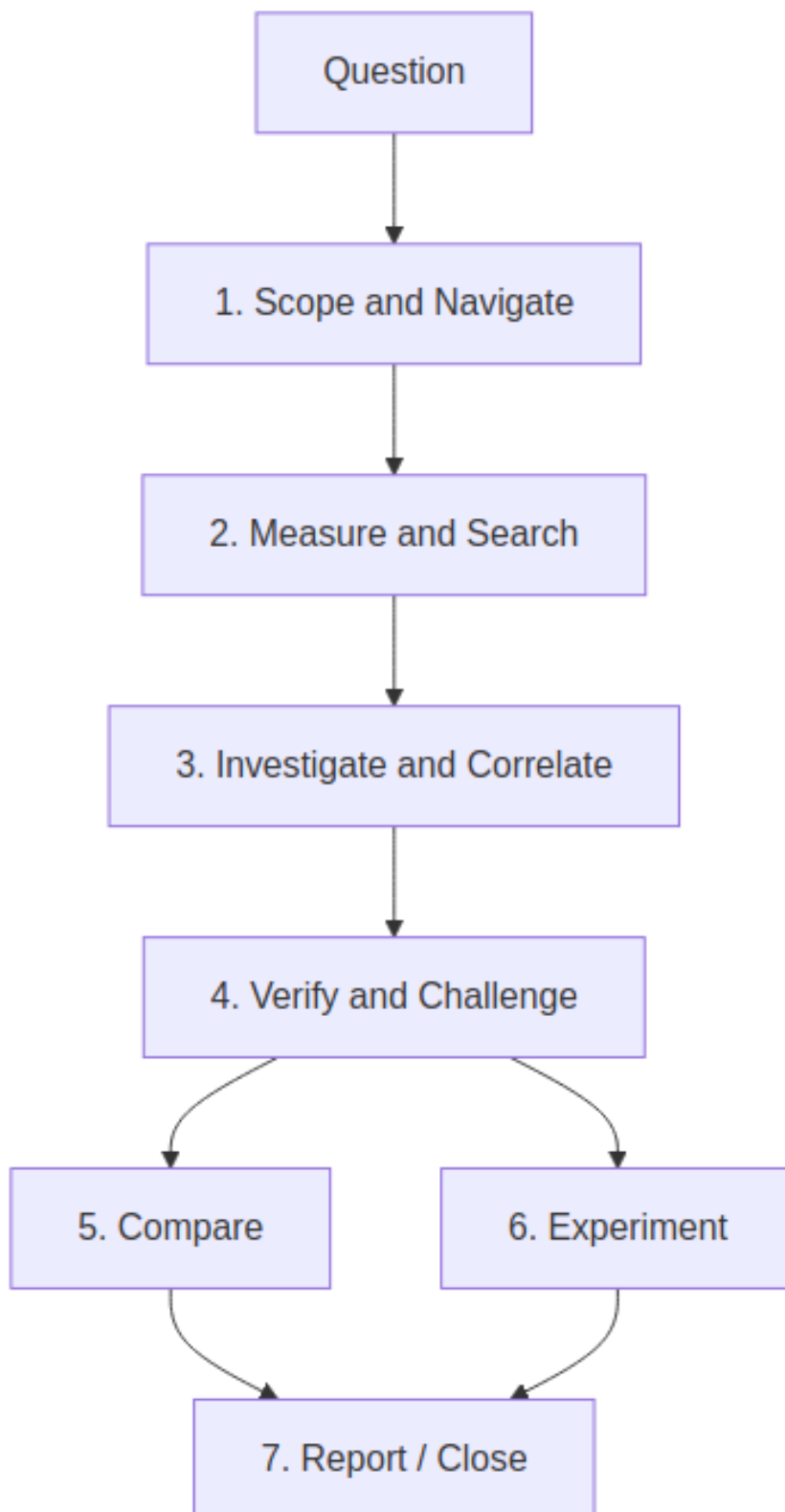
Read-only evidence tools and export tools (export_report, export_investigation) run immediately; GUI-changing tools wait for **Apply** unless **Auto-apply GUI actions** is on. Names and parameters are in [Complete GUI tool reference](#) below.

It is easier to understand the AI tools by **purpose** than by function name.

Most users should start with the built-in templates and the Investigation plan.

The individual tool schema is mainly for advanced use, debugging, and implementation reference.

Tool mental model



Apply, Skip, and Undo

Tools fall into two behavioral classes:

Class	Behavior	Examples
Read-only evidence tools	Run immediately; they do not change the viewer	query_raw_metric, search_timeline, investigate, correlate_events, find_critical_path, verify_claim
GUI-changing tools	With Auto-apply GUI actions off (default), the batch waits for Apply / Skip	set_cursors, zoom_to_range, highlight_task, set_view_mode, add_annotation, bookmark_finding

Several tool calls may arrive in one model turn and are applied as one batch. **Undo last actions** restores zoom / view / highlight / inspector / marks; Ctrl/Cmd+Z also reverts cursors and marks. Export tools still open a save dialog.

1. Scope & navigate — “Where should I look?”

Use these tools to turn an answer into a visible timeline location.

Goal	Main tools	Result
Scope a suspected phase	set_cursors, zoom_to_range	Places C1-Cn and focuses the relevant interval
Focus a task	highlight_task	Lock-highlights the task on the timeline
Change perspective	set_view_mode	Task/Core and horizontal/vertical view
Inspect a migration corridor	open_corridor_inspector	Opens the Migration & Corridor Inspector

Goal	Main tools	Result
Preserve evidence	bookmark_finding, add_annotation	Adds semantic or free-text timeline marks
Clean up	clear_marks, reset_view	Clears investigation clutter or restores full-span view

Beginner usage: normally accept these as GUI cards produced by **Investigate**, **Verify**, or **Explain region** rather than calling them directly.

2. Measure & search — “What happened?”

These tools fetch deterministic evidence. They do not change the trace.

Question	Main tools	Evidence returned
Where did an event occur?	search_timeline	Matching task/STI/tag/interval/pointer/migration timestamps
What are this task’s measured values?	query_raw_metric	Scoped execution, blocking, migration, sync, PI, or findings rows
Is the distribution unusual?	analyze_distribution	p50-p99.9, standard deviation, CV, outlier rate
Is timing periodic or jittery?	analyze_periodicity	Expected vs observed period/jitter statistics
Does the task exceed a budget?	check_budget	WCET/response/deadline budget comparison

These tools are the evidence layer. They should be preferred over asking the model to guess a number.

3. Investigate & correlate — “What is related?”

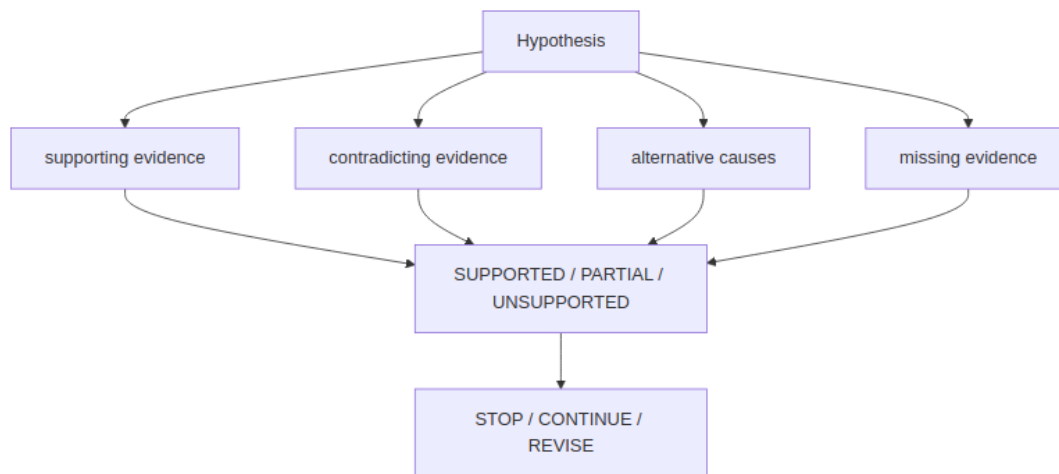
Use this group after Triage identifies a concrete issue.

Depth	Main tools	Purpose
Triage	<code>detect_anomalies</code> , <code>cluster_findings</code> , <code>cluster_incidents</code>	Rank and group Findings
Investigate	<code>investigate</code> , <code>plan_investigation</code> , <code>suggest_scope</code>	Build hypotheses and choose the cheapest next checks
Correlate	<code>correlate_events</code> , <code>find_related_findings</code>	Merge nearby execution/blocking/migration/sync/priority evidence
Path	<code>find_critical_path</code>	Walk preemption, blocking, and mutex activity around an incident
Dependency	<code>build_task_dependency_graph</code>	Show wait/preempt/migrate/PI relationships
Temporal	<code>analyze_temporal_causality</code>	Build a happens-before chain from observed evidence times
Root cause	<code>build_causal_chain</code> , <code>rank_root_causes</code>	Rank hypotheses while labeling causal/correlated/temporal relationships

Investigate / Root cause / Auto investigate orchestrate these deeper tools. Most users should not need to choose the sequence manually.

4. Verify & challenge — “Is this really the cause?”

This group prevents a plausible story from being treated as a confirmed diagnosis.



Purpose	Main tools
Check one claim	<code>verify_claim</code>
Find evidence against a hypothesis	<code>detect_contradictions</code>
Decide whether enough evidence has been gathered	<code>assess_evidence_sufficiency</code>
Force alternative explanations	<code>challenge_conclusion</code>
Track hypothesis state	<code>manage_hypotheses</code>
Inspect priority-inversion evidence	<code>detect_priority_inversion</code>
Evaluate investigation quality	<code>score_investigation</code>

The Evidence & Validation panel complements these tools with status, a direct-evidence table, checks, missing evidence, next action, and investigation details (quality, cost, trees).

5. Compare — “What changed?”

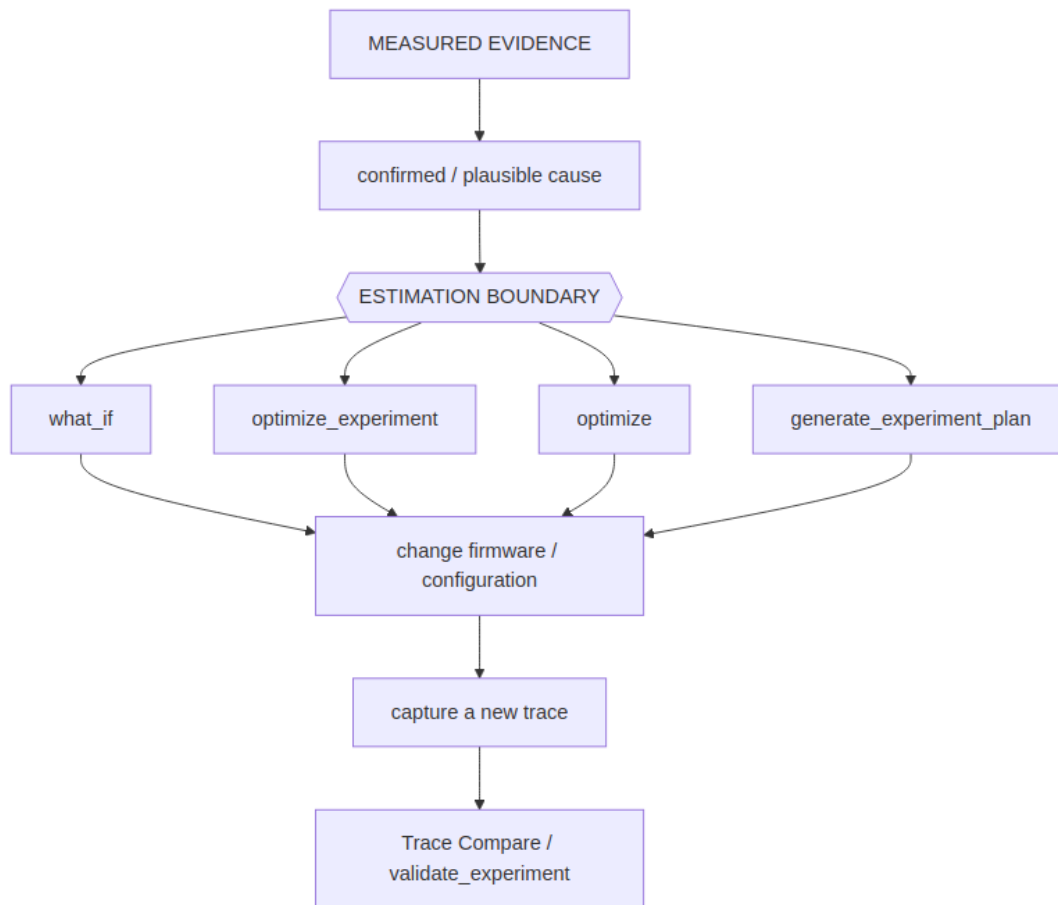
Use these tools when two or more traces are open.

Goal	Main tools	Result
Open / obtain Trace Compare	trigger_ compare	Same comparison data as toolbar Compare
Compare two builds	compare_ performance	Structured metric deltas + confidence
Explain the primary regression	regression_ explain	Narrative tied to A/B deltas
Localize the regression	regression_ localize	Suspect task / region / mechanism
Compare two tasks	compare_tasks	Side-by-side task metrics
Compare against history	baseline_ score	Drift against a stored baseline
Rank several open traces	analyze_ traces	Relative scheduling behavior

A comparison is only meaningful when the two traces represent equivalent workload phases.

6. Experiment — “What should I try?”

These tools operate **after** a cause has enough evidence.



Goal	Main tools
Test one concrete idea	<code>what_if</code>
Rank candidate changes	<code>optimize_experiment</code>
Get qualitative mitigation ideas	<code>optimize</code>
Generate bench/firmware validation steps	<code>recommend_experiments</code> , <code>generate_experiment_plan</code>
Compare prediction with measured result	<code>validate_experiment</code>
Store the outcome	<code>record_experiment_outcome</code>

Important: `what_if` and optimization results are estimates, not measured scheduler behavior.

7. Report & close — “What did we learn?”

Goal	Main tools
Generate structured engineering text	<code>generate_report</code>
Save a diagnostic report	<code>export_report</code> (HTML summary first; transcript in appendix)
Save a complete Investigation Case	<code>export_investigation</code>
Replay/summarize the investigation	<code>investigation_replay</code> , <code>summarize_investigation_context</code>
Remember similar cases	<code>investigation_memory</code> , <code>find_similar_investigations</code>
Finish the Case	<code>close_investigation</code>

HTML `export_report` builds a **diagnostic report**: executive summary (status + completeness), coverage, ranked findings, observation vs interpretation, in-scope evidence table, and next action. Conversation, GUI state, out-of-window evidence, and metadata sit in an expandable **Appendix**. Export runs immediately (query/export batches need no Apply); if a turn is still in flight the HTML marks **Analysis incomplete**. Optional mode: `summary` (default), `technical`, or `full` (appendix conversation open).

Complete GUI tool reference

The table below is the exhaustive schema reference. Use it when implementing, debugging, or explicitly steering tool calls.

Tool	Parameters / targets	Effect
set_cursors	timestamps (1-8 trace times)	Place cursors (enables Limit to C1-Cn when two or more)
zoom_to_range	start_time, end_time	Focus the timeline between two times
highlight_task	task_name_or_id (display name, numeric id, or merge key)	Lock-highlight a task row. Unknown names are ignored so the timeline is not dimmed. Empty string clears.
set_view_mode	mode (task / core); optional orientation	Switch Task or Core view; horizontal or vertical
open_corridor_inspector	optional core_from / core_to (Core_0, 0, c0, Core 0)	Open Migration Inspector; aliases resolve the same way
add_annotation	time, note (≤240 chars)	Pin an orange timeline note at a timestamp (stays on the current right-panel tab)
query_raw_metric	task, metric (priority_inheritance, execution, migrations, blocking, sync, findings)	Read-only: return the per-task series for the current Statistics scope (up to 40 rows)

Tool	Parameters / targets	Effect
export_report	optional format (html / csv / json), optional mode (summary / technical / full)	HTML diagnostic report: executive summary, coverage, ranked findings, in-scope evidence, next action; conversation/GUI/rejected evidence in <details> appendix. Runs immediately (no Apply). Mid-flight exports still download and mark Analysis incomplete in the HTML. Strips export_report tool cards from the transcript. json saves a full investigation package (see export_investigation).
clear_marks	optional what (annotations / cursors / bookmarks / all / everything)	Clear AI clutter. all (default) drops annotations + cursors; everything also clears bookmarks
reset_view	(none)	Fit the timeline to the full span and clear the task highlight (marks stay)

Tool	Parameters / targets	Effect
search_timeline	query; optional mode (contains / exact / regex / sti / tags / intervals / lifecycle / pointers / migrations)	Find-panel search; returns matching timestamps (up to 40)
trigger_compare	optional tab_a / tab_b (0-based tab index or filename)	Read-only Trace Compare CSV + open the same dialog as toolbar Compare (needs two loaded tabs)
investigate	optional finding_id, depth (1-5)	Read-only: investigation graph with root-cause chain, hypotheses, ranked anomalies, suggested tools
detect_anomalies	optional limit (1-40)	Read-only: rank Analysis Findings as Critical / Warning / Info
correlate_events	task; optional around_time, window	Read-only: merge blocking / execution / migration / sync / priority / Find hits into one timeline

Tool	Parameters / targets	Effect
find_critical_path	task; optional timestamp, window (default 2000)	Read-only: preempt/block/mutex critical path around a timestamp; also returns a mermaid graph (graph LR), graph_nodes (id/label/kind/time), and split blocking_ steps / preemption_ steps arrays. Path steps carry start/stop; Evidence bullets use clickable range: L0/HI to zoom and place C1-C2 on the episode
compare_performance	optional tab_a / tab_b	Read-only: structured A vs B metric deltas + confidence (two tabs); data.regression_ type classifies the primary delta as execution / scheduling / synchronization / migration / load_ balance / unknown (legacy classification values like thrashing / load_imbalance / tick_health are preserved alongside it)

Tool	Parameters / targets	Effect
generate_report	optional report_type, finding_id	Read-only: typed engineering markdown (executive / performance / root_cause / regression / optimization / bug / ci); call export_report to save
check_budget	optional budgets, tasks	Read-only: compare per-task WCET/response/deadline metrics against budgets (host builds rows from findings when tasks omitted)
optimize	optional limit (default 5)	Read-only: evidence-backed mitigation ideas (estimate disclaimer)
regression_explain	optional tab_a / tab_b	Read-only: compare two tabs then narrate the primary regression; includes the same regression_type classification
bookmark_finding	time, kind (root_cause / evidence / correlated / reference); optional note	GUI: pin a semantic investigation annotation (Apply required)
investigation_replay	optional finding_id, conclusion, tools_run, evidence_times	Read-only: structured investigation replay card

Tool	Parameters / targets	Effect
what_if	change; optional task	Read-only: heuristic slice-replay what-if (migrations / blocking / load balance; not an RTOS kernel)
optimize_experiment	optional task, limit (1-12, default 5)	Read-only: run ranked automatic pin/priority/contention/migration experiments
analyze_traces	(none)	Read-only: rank all loaded tabs by scheduling behavior
baseline_score	optional task, baseline, snapshot	Read-only: score current per-task metrics (WCET/blocking/migrations/response) against the stored historical baseline; flags $ z > 2$
recommend_experiments	optional finding_id, task, limit (1-20, default 5)	Read-only: suggest simulation / firmware / measurement validation experiments from findings heuristics
export_investigation	optional finding_id, conclusion, tools_run, evidence_times	Download the completed investigation (finding, tools run, queries, evidence, conclusion, confidence, alternatives) as a JSON package

Tool	Parameters / targets	Effect
detect_priority_inversion	optional task, window	Read-only: scan priority-inheritance boost episodes for L/M/H inversion suspects (high/medium/low task, mutex, time, duration)
find_related_findings	optional finding_id, task, metric, window, limit (1-40, default 10)	Read-only: relate Analysis Findings by shared task, metric keyword, evidence-time proximity, or severity adjacency
compare_tasks	task_a, task_b; optional metrics	Read-only: side-by-side execution/blocking/migrations/priori inheritance delta table between two tasks
explain_finding	optional finding_id, level (quick / technical / deep)	Read-only: explain one Analysis Finding at the chosen depth (host-side; uses finding text plus hypotheses)
interpret_query	question	Read-only: turn a free-form question into an explicit investigation mode/scope before other tools run

Tool	Parameters / targets	Effect
validate_experiment	optional expected, actual (metric → signed percent)	Read-only: compare expected experiment deltas with actual A vs B / what-if results (VALIDATED / PARTIALLY VALIDATED / DISPROVED)
manage_hypotheses	hypothesis_id, status (supported / possible / rejected / need_evidence); optional reason, finding_id	Read-only: mark one investigation hypothesis status
plan_investigation	optional question, finding_id	Read-only: rank hypotheses and the cheapest tool sequence
suggest_scope	optional question	Read-only: recommend task / related tasks / time window
detect_contradictions	optional hypothesis, metrics	Read-only: verdict SUPPORTED / CONTRADICTED / INSUFFICIENT
assess_evidence_sufficiency	optional tools_run	Read-only: STOP INVESTIGATION / CONTINUE / REVISE HYPOTHESIS
cluster_findings	(none)	Read-only: group related findings into incidents

Tool	Parameters / targets	Effect
generate_fingerprint	(none)	Read-only: HIGH/MEDIUM/LOW scheduling, sync, and timing bands
find_similar_ investigations	optional limit	Read-only: match fingerprint against recorded experiment outcomes
regression_localize	optional label_a, label_b	Read-only: localize A vs B inflation to a task and region
build_causal_chain	(none)	Read-only: causal / correlated / temporal edges (never silent causation)
generate_experiment_plan	optional task, limit	Read-only: ranked firmware / what-if experiments
record_experiment_outcome	optional change, predicted, actual, quality	Read-only: store outcome for later similar-case matching
score_investigation	optional tools_run, conclusion, confidence, elapsed_ s	Read-only: evidence efficiency, cost, false-confidence, falsification, scope, stop
analyze_temporal_causality	optional task	Read-only: happens-before chain from Findings times

Tool	Parameters / targets	Effect
build_task_dependency_graph	optional task	Read-only: BTF wait/preempt/migrate/PI graph; 2-hop neighborhood + upstream tasks
decompose_response_time	optional task	Read-only: relative delay-component shares
rank_root_causes	(none)	Read-only: rank causes from findings/hypotheses
verify_claim	claim; optional claim_type, subject, object, evidence	Read-only: SUPPORTED / PARTIAL / UNSUPPORTED
challenge_conclusion	optional conclusion	Read-only: alternatives and missing evidence
investigation_memory	optional action (recall / store), record, limit	Read-only: persist/recall similar cases
cluster_incidents	optional window_ns	Read-only: time-proximity incident clusters
close_investigation	optional conclusion, confidence	Read-only: close the case envelope
analyze_distribution	optional values, metric (auto / execution / blocking / priority_inheritance / tick), task	Read-only: p50/p90/p95/p99/p99.9, stddev, CV, 3-sigma outlier rate. Statistics Query with AI... on a distribution chart harvests the open plot's samples.

Tool	Parameters / targets	Effect
analyze_periodicity	optional times, expected, source (auto / tick / sti / isr / timer / release), task, durations	Read-only: expected vs p50/p99/max, RMS and peak-to-peak jitter, kind
summarize_investigation_context	optional conclusion, tools_run	Read-only: compact investigation snapshot

Models without native tool calling can emit a fenced `btftool` JSON block; the viewer renders the same GUI cards. Prefer a tool-capable model for investigation-heavy workflows.

Desktop and web behavior

BTFViewer Desktop and Web are intended to provide the **same AI investigation workflow, tool behavior, evidence model, and validation rules**. For normal use, there is no separate “Desktop workflow” or “Web workflow” to learn.

Use whichever frontend fits your environment:

- **Desktop** — convenient for local files, native save dialogs, and local/private AI endpoints.
- **Web** — convenient when you want a browser-only viewer or a hosted/development deployment.

The differences are mostly platform integration details rather than AI capabilities.

Platform detail	Desktop	Web
AI tools, Investigation Case, Evidence panel, validator	Same behavior	Same behavior
Task/event/region AI actions	Same behavior	Same behavior

Platform detail	Desktop	Web
Model picker and endpoint configuration	Supported	Supported
Reports / investigation export	Native save dialog	Browser download
In-chat diagrams	Rendered in the desktop UI	Rendered as inline browser content
Self-signed HTTPS endpoint	Can optionally allow self-signed TLS per preset	Browser/OS certificate policy still applies
Local file:// launch	Not applicable	Cross-origin requests may be blocked; use the development/preview server when needed

User takeaway: AI analysis should produce the same investigation and evidence on Desktop and Web. Platform-specific differences matter mainly when configuring endpoints, certificates, downloads, or browser networking.

Detailed platform-specific setup problems are documented in **Troubleshooting** below rather than treated as separate AI features.

Troubleshooting

Symptom	Cause	Try
Web: Failed to fetch / CORS	Browser blocked a cross-origin call (file:// sends Origin: null)	Prefer <code>npm run dev / make preview</code> (both proxy Ollama), or see Opening the web app from file://

Symptom	Cause	Try
401 / 403	Missing or rejected key / origin	Settings → AI → Sign in or API key (OPENAI_API_KEY / GEMINI_API_KEY / OLLAMA_API_KEY; local Ollama needs none)
CERTIFICATE_VERIFY_FAILED / self-signed TLS	Private CA or self-signed HTTPS gateway	Desktop: Settings → AI → Allow self-signed TLS . Web: trust the cert in the OS/browser, use http:// on a private LAN, or use the Desktop app
Chat probe timed out / The read operation timed out	GET /models lists ids only; inference is slow or hung	Test connection POSTs /chat/completions (non-streaming, 120s). Warm the model (ollama run MODEL) and retry. Debug with the curl probe below; if curl hangs too, the gateway's chat upstream is stuck. Try "stream": true if non-stream never returns. Lower context length on a VRAM-tight local host.
Model not found	Typed id is not served	Refresh the Model list (or Test connection) and pick a served id from the dropdown, or ollama pull it

Symptom	Cause	Try
Gemini HTTP 400 thought_signature	Gemini 3 requires a thought blob on tool follow-ups	Retry the question — the viewer echoes Gemini thought signatures
Gemini HTTP 400 function_response.name	OpenAI-compatible follow-up with empty tool_calls[].id	The viewer fills ids and role=tool names before the next turn. Retry the case.
Raw btftool JSON instead of native tool calls	Model lacks or skips function calling	The viewer renders the same cards. Select Apply or enable Auto-apply GUI actions . For reliable native calls, use a tool-capable model such as qwen3.5:9b or a supported cloud model.
Ask times out (over 120s) or stays on Waiting...	Cold start, CPU offload, or VRAM spill	Stop (composer icon), warm with ollama run MODEL, retry. Use Clear between long threads. Smaller model or shorter Statistics scope if the Findings card is huge
Later turns ignore earlier facts	Chat history exceeded the context window	Clear on the AI bar, or Analysis → Query with AI... / toolbar Compare → Query with AI... for a fresh scoped prompt

Symptom	Cause	Try
Need raw AI request/response dumps (Desktop)	Debugging tool rounds / provider quirks	Settings → AI → Log MCP messages to file (off by default). Appends to <code>./ai_mcp_messages.log</code> ; delete when finished

Test connection curl

Same body the viewer sends for **Test connection** (replace BASE, MODEL, and KEY):

```
curl -vk --max-time 180 \  
  -H "Authorization: Bearer KEY" \  
  -H "Content-Type: application/json" \  
  -d \  
    '{ "model": "MODEL", "stream": false, "messages": [{ "role": "user", "content": "Reply with JSON only: {\"ok\": true}\" }], "max_tokens": 24 }' \  
  BASE/chat/completions
```

Opening the web app from file://

A page opened straight from disk sends `Origin: null`, which Ollama rejects with 403 — the browser then reports only `Failed to fetch`. Serving the app over http avoids this entirely (`npm run dev / make preview proxy Ollama for you`), and the Desktop app is not affected at all.

To keep using `file://`, allow every origin on the Ollama side:

```
# Server started from a terminal  
OLLAMA_ORIGINS="*" ollama serve
```

```
# macOS menu-bar app (Ollama.app) – a shell variable does not reach it  
launchctl setenv OLLAMA_ORIGINS "*" # then quit Ollama and reopen it
```

Verify the change took effect; expect 200 and an `Access-Control-Allow-Origin` header:

```
curl -s -D - -o /dev/null -H "Origin: null" http://localhost:11434/v1/models
↪ \
  | grep -iE "^HTTP|access-control-allow-origin"
```

If a file:// page is still refused, list the null origin explicitly with `OLLAMA_ORIGINS="*,null"`. Note that * lets **any** page you visit reach your local models; undo it with `launchctl unsetenv OLLAMA_ORIGINS` when done.

CLI regression gate

Desktop headless CI can compare a candidate trace to a baseline and optionally ask the configured AI for a short narrative:

Headless analyze with `--fail-on-regression`:

```
python builds/btf_viewer.py analyze candidate.btf --baseline baseline.btf
↪ --fail-on-regression
python builds/btf_viewer.py analyze candidate.btf --save-baseline
↪ /tmp/base.json
python builds/btf_viewer.py analyze candidate.btf --baseline /tmp/base.json
↪ --fail-on-regression --ai
python builds/btf_viewer.py ai-test --dataset tests/ai --fail-under 70
python builds/btf_viewer.py ai-test --config examples/ai/benchmark.xml -o
↪ AI_BENCHMARK.md
python builds/btf_viewer.py ai-test --config examples/ai/benchmark.xml
↪ --compare-context -o AI_BENCHMARK.md
python builds/btf_viewer.py ai-test --config examples/ai/benchmark.xml
↪ --insecure
```

Or make `-C BTFViewer ai-test (AI_DATASET, AI_FAIL_UNDER)`, make `-C BTFViewer ai-test-live (AI_CONFIG, optional AI_MODELS, writes AI\_BENCHMARK.md)`, and make `-C BTFViewer ai-test-context` (same as live + `--compare-context`). Dataset, scoring rules, and context-mode flags: [Benchmark / evaluation suite](#).

See also [Export → Headless CLI](#) in the user guide.

Benchmark and evaluation suite

Offline `ai-test / runOfflineBenchmark` already ships. Live runs read **model id, base URL, TLS, and API key** from a suite XML (`--config examples/ai/benchmark.xml`) and write `AI_BENCHMARK.md` — a rerun **merges** into an existing file (untouched models/context-modes/cases are preserved byte-for-byte; `--replace-report` overwrites fully). Commands: **CLI regression gate**. Default live scoring uses **Full evidence** (`--context-mode full`). **--compare-context** runs Compact, Balanced, and Full on the same cases and reports score, token totals, and latency side by side. Each live model call **retries up to 10 times, pausing 10s between attempts** on transient errors (HTTP 429/503 high demand, timeouts, empty replies); auth and not-found errors are not retried.

The capability matrix above is qualitative (small local vs 9B+ vs cloud). The suite turns those expectations into repeatable measurements: **which model is most reliable for BTF Viewer trace investigation**, not which model is largest or “smartest.”

Context mode benchmarking

Live `ai-test` uses the same Compact / Balanced / Full evidence packing as **Settings → AI → Context** (Findings trim, stage-filtered tool schemas, Compact reply cap). Settings in the GUI do not affect the CLI scorer.

Flag	Purpose
<i>(default)</i>	Full evidence — complete Findings and tool catalog
<code>--context-mode compact</code>	Single run in Compact (or balanced, full; comma-separated for a subset)
<code>--compare-context</code>	Run all three modes per model on the same cases

Each live case records **overall score, pass/fail, prompt / completion / total tokens** (summed across tool follow-ups), and **elapsed time**. With `--compare-context`, `AI_BENCHMARK.md` adds a **Context mode comparison** table per model (score vs tokens vs mean latency).

```
python builds/btf_viewer.py ai-test -c examples/ai/benchmark.xml
  ↪ --compare-context -o AI_BENCHMARK.md
make -C BTFViewer ai-test-context # AI_CONFIG, optional AI_MODELS
```

Use this when choosing a default Context setting. Compact used fewer tokens for every measured model, but it did not always reduce latency or preserve the score. Compare all three modes on the intended endpoint and workload. Details: [Context mode \(token usage\)](#).

Scope

Keep the live set focused on:

- **Gemini cloud models**
- **Local Ollama models that are practical on a typical developer workstation**

Do **not** pick local models only because they are newest or largest. Measure models that can run alongside BTF Viewer, Ollama, and the AI context/tooling workload.

Recommended models

Local — developer workstation:

- **Qwen3.5 9B** (qwen3.5:9b) — shipped in-app default and the practical local choice. Balanced passed 15/17 cases; Full evidence produced Overall 88.
- **Qwen3.8 27B** (qwen3.8:27b) — high-latency local comparison. Balanced reached Overall 88, but averaged 325.2s/case and did not consistently outperform the 9B model.

Gemini (configurable; newer ids can be added without changing the runner):

- **Gemini 3.7 Flash** (gemini-3.7-flash) — higher-scoring shipped Gemini reference. Full evidence reached Overall 85 with 14/17 PASS.
- **Gemini 3.5 Flash-Lite** (gemini-3.5-flash-lite) — latency-focused cloud reference. Full evidence reached Overall 83 at 2.6s/case.

Shipped live suite

```
|
|— Local
|   |— Qwen3.5 9B
```

```
|   └─ Qwen3.8 27B
|
└─ Gemini
    ├── Gemini 3.7 Flash
    └─ Gemini 3.5 Flash-Lite
```

The recorded results show why model size alone is not a useful selection rule. The 27B local model matched the highest local Overall score only in Balanced mode, while taking about 325 seconds per case. The 9B model reached Overall 88 in Full evidence at 16.2 seconds per case. Measure both **diagnostic quality** and **practical system performance**.

Do not hard-code the model list into the runner. Copy [examples/ai/benchmark.xml](#). For a self-signed or private-CA gateway, keep `<tls-verify>false</tls-verify>` (the suite default); public HTTPS models can override to true:

```
<ai-benchmark version="1">
  <dataset>tests/ai</dataset>
  <fail-under>0</fail-under>
  <output>AI_BENCHMARK.md</output>
  <endpoint>
    <base-url>http://localhost:11434/v1</base-url>
    <tls-verify>false</tls-verify>
    <timeout-s>360</timeout-s>
  </endpoint>
  <models>
    <model id="qwen3.5:9b"/>
    <model id="qwen3.8:27b"/>
    <model id="gemini-3.7-flash" preset="gemini">
      <base-
⇨ url>https://generativelanguage.googleapis.com/v1beta/openai</base-url>
      <tls-verify>true</tls-verify>
      <api-key env="GEMINI_API_KEY"/>
    </model>
    <model id="gemini-3.5-flash-lite" preset="gemini">
      <base-
⇨ url>https://generativelanguage.googleapis.com/v1beta/openai</base-url>
      <tls-verify>true</tls-verify>
      <api-key env="GEMINI_API_KEY"/>
    </model>
  </models>
</ai-benchmark>

<!-- Self-signed / private CA gateway -->
```

```
<endpoint>
  <base-url>https://llm.internal.example:8443/v1</base-url>
  <tls-verify>>false</tls-verify>
  <api-key env="GATEWAY_API_KEY"/>
</endpoint>
```

<api-key env="VAR"> reads the environment first, then any text inside the element. Omit the text (and do not commit secrets). `tls-verify false`, or `ai-test --insecure`, skips certificate checks on Desktop. `--models id1,id2` (or make `ai-test-context AI_MODELS=id1,id2`) selects a subset of <model> entries. A custom suite may mark models `optional="true"` so they are skipped when their API key is missing unless you name them in `--models / AI_MODELS`. For Ollama, list the ids you actually have pulled. Record the exact model identifier and runtime configuration.

`--only-cases id1,id2` (or make `ai-test-context AI_CASES=id1,id2`) restricts scoring to specific tests/ai dataset case ids — handy for re-testing a few cases that returned ERROR (transient HTTP 429/503) without rerunning the whole suite.

When `-o/--output` already exists, `ai-test` **merges** this run into it instead of overwriting: any model/context-mode block or offline case that was actually rerun is replaced, and every other block/case already in the file is left byte-for-byte untouched (the Comparison, Context mode comparison, and Metric breakdown tables are re-computed from the merged set). This makes narrow reruns safe, e.g. re-scoring just one model in one context mode:

```
python builds/btf_viewer.py ai-test -c examples/ai/benchmark.xml \
  --models gemini-3.7-flash --context-mode full -o AI_BENCHMARK.md
make -C BTFViewer ai-test-live AI_MODELS=gemini-3.7-flash AI_CONTEXT=full
```

Pass `--replace-report` (or `AI_REPLACE=1`) to overwrite the file fully instead of merging — useful for a fresh, full-suite run you want to replace stale results wholesale.

In-app picker (not shipped): **Settings** → **AI** → **Benchmark** with checkboxes for Gemini and local Ollama models, then **Run Benchmark**.

Dataset

`tests/ai/` holds known traces for the major diagnostic scenarios. Each case has **expected facts**, not an exact natural-language answer:

`tests/ai/`

- |— migration_thrash.btf
- |— mutex_contention.btf
- |— priority_inversion.btf
- |— deadline_miss.btf
- |— load_imbalance.btf
- |— trace_regression.btf
- |— explain_region.btf
- |— adversarial_mutex_vs_starvation.btf
- |— adversarial_exec_vs_preemption.btf
- |— adversarial_correlation_not_cause.btf
- |— adversarial_out_of_scope_time.btf
- |— period_jitter.btf
- |— waiter_owner_handoff.btf
- |— stats_page_next_check.btf
- |— response_vs_blocking.btf
- |— preempt_matrix_vs_chain.btf
- |— mutex_block_vs_wait_queue.btf

Adversarial cases (kind adversarial) use a decoy finding or timestamp. The obvious answer is wrong:

Case	Decoy	Actual
adversarial_mutex_vs_starvation	mutex contention	CPU starvation / preemption
adversarial_exec_vs_preemption	long execution / WCET	preemption
adversarial_correlation_not_cause	ISR caused Comm latency	correlation, no causal link
adversarial_out_of_scope_time	diagnose jump:9000	timestamp outside the cursor window
period_jitter	tick health / tickless	task inter-arrival; open Period / Jitter
waiter_owner_handoff	kernel wait-queue	heuristic mutex handoff; open Waiter x Owner
stats_page_next_check	invent detect_timeline_anomalies	open Timeline Anomalies / Worst Events

Case	Decoy	Actual
response_vs_blocking	Blocking Time is end-to-end response	open Response Time
preempt_matrix_vs_chain	invent detect_preemption_matrix	open Preemption Matrix
mutex_block_vs_wait_queue	reconstruct the kernel wait queue	open Mutex Blocking

```
id: migration_thrash
trace: migration_thrash.btf
expected:
  finding_types: [migration, load_balance]
  tasks: [CS[22]]
  evidence:
    required_metrics: [migrations]
  allowed_tools: [detect_anomalies, correlate_events, query_raw_metric]
  forbidden:
    invented_task_names: true
    out_of_scope_timestamps: true
```

That keeps scoring robust against harmless wording differences.

Evaluation metrics

Metric	What it measures
Finding identification	Did the model identify the expected problem?
Evidence accuracy	Are cited metrics/events actually present? required_metrics also accepts Statistics page titles (Period / Jitter, Waiter × Owner, Timeline Anomalies, ...) and common aliases (Period/Jitter, Waiter x Owner)
Timestamp validity	Are jump:TIME values real and in scope?
Task-name validity	Did the model use only known task names?
Tool selection	Did it call appropriate investigation tools?

Metric	What it measures
Tool-chain quality	Did it gather enough evidence before concluding?
Root-cause accuracy	Does the conclusion match the expected diagnosis?
Alternative handling	Did it consider plausible alternatives?
Confidence calibration	Is confidence consistent with the available evidence?
Response completeness	Did it answer the investigation question completely?
Latency	How long did the investigation take?
Tool-call count	How many tool rounds were required?
Peak memory	How much RAM was consumed during inference?
Time to first token (TTFT)	How quickly did the model begin responding?
Generation throughput	Sustained tokens/sec during the investigation
Investigation success rate	Percentage of cases completed correctly within the configured time/resource limit
False-causal rate	Claimed a causal link the case marks as coincidence / non-causal (0-100, higher is worse)
False-confirmation rate	Confirmed the decoy finding (trap_phrases) instead of the real cause
Unsupported-claim rate	Share of validator claims that fail task/time/scope checks
Premature-conclusion rate	High confidence or a conclusion before required tools ran

For local runs, memory and latency are first-class. A slightly more accurate model that is unusable under memory pressure should not automatically rank higher.

Level 1 — tool / evidence correctness: valid tool, parameters, task, timestamp, and scope. Isolates tool-use bugs from reasoning quality.

Level 2 — diagnostic correctness: expected vs actual diagnosis, evidence, and alternatives. A convincing explanation is not enough.

Headline score: weighted engineering score (Finding / Evidence / Tool use / Root cause / Calibration / Safety), not a probability of correctness. Keep the component

scores visible. PASS is overall ≥ 70 .

Safeguards the suite must test: no invented task names, metric values, or jump:TIME; no timestamps outside the cursor region; no unsupported conclusions presented as confirmed; evidence must match tool results; heuristic what-if stays labeled as estimates; the model must not claim a simulation is a measured result.

Model matrix

Same suite against the shipped Gemini and local Ollama models. Recorded 2026-08-19 (17-case dataset); full case tables and Compact/Balanced numbers: [AI_BENCHMARK.md](#). Scores below are the **Full evidence** context mode (the live-scoring default).

Model	Category	Finding	Root cause			Calibration	Notes
			Evidence	Case	Tool		
qwen3.5:9b	Local / practical	85	93	82		80	overall 88 , 16.2s/case, 14/17 PASS
qwen3.8:27b	Local / high-latency	88	94	65		80	overall 86 , 332s/case, 13/17 PASS
gemini-3.5-flash-lite	Cloud / fast	82	90	71		80	overall 83 , 2.6s/case, 13/17 PASS
gemini-3.7-flash	Cloud	82	94	59		80	overall 85 , 25.0s/case, 14/17 PASS

[AI_BENCHMARK.md](#) also carries two cloud models run outside the shipped suite (--models against a private config), for reference only — they are not in examples/ai/benchmark.xml and are not part of the “Recommended models” guidance below:

Model	Category	Root Cause			Calibration	Notes
		Finding	Evidence	cause		
claude-sonnet-5	Cloud (optional, not shipped)	85	94	59	80	overall 82 , 15.9s/case, 12/17 PASS
gpt-5.6-sol	Cloud (optional, not shipped)	91	90	76	80	overall 88 , 9.6s/case, 15/17 PASS

Live --config runs a tool-result follow-up when the first turn is tools-only (or planning text without a Confidence line). Single-turn scores are not comparable.

The results support these practical conclusions:

- **Best practical local setup:** qwen3.5:9b with Balanced for the highest pass count, or Full evidence for the highest Overall score.
- **Fastest measured cloud setup:** gemini-3.5-flash-lite with Compact at 2.3s/case; Full evidence improved its score to 83 at 2.6s/case.
- **Best shipped Gemini result:** gemini-3.7-flash with Full evidence, Overall 85 and 14/17 PASS.
- **Highest result in the complete report:** optional gpt-5.6-sol with Compact, Overall 90 and 16/17 PASS.
- **More context is not automatically better:** the best mode depends on the model. Compare all three modes before choosing a deployment default.

Context mode comparison (--compare-context): runs the same live suite three times per model (Compact → Balanced → Full evidence) and writes a **Context mode comparison** table with overall score, pass rate, prompt/completion/total tokens, and mean latency. Use this to see whether a smaller context budget saves tokens/time without hurting investigation scores on your hardware.

Context-size (Findings + tools + history). Do not judge a local model only by tokens/sec — check tool use and grounding as context grows:

Context	Purpose
8K	Minimum investigation workload
16K	Typical investigation

Context	Purpose
32K	Large Findings / multi-tool investigation
64K	Stress test, if supported

Practical comparison on a developer workstation:

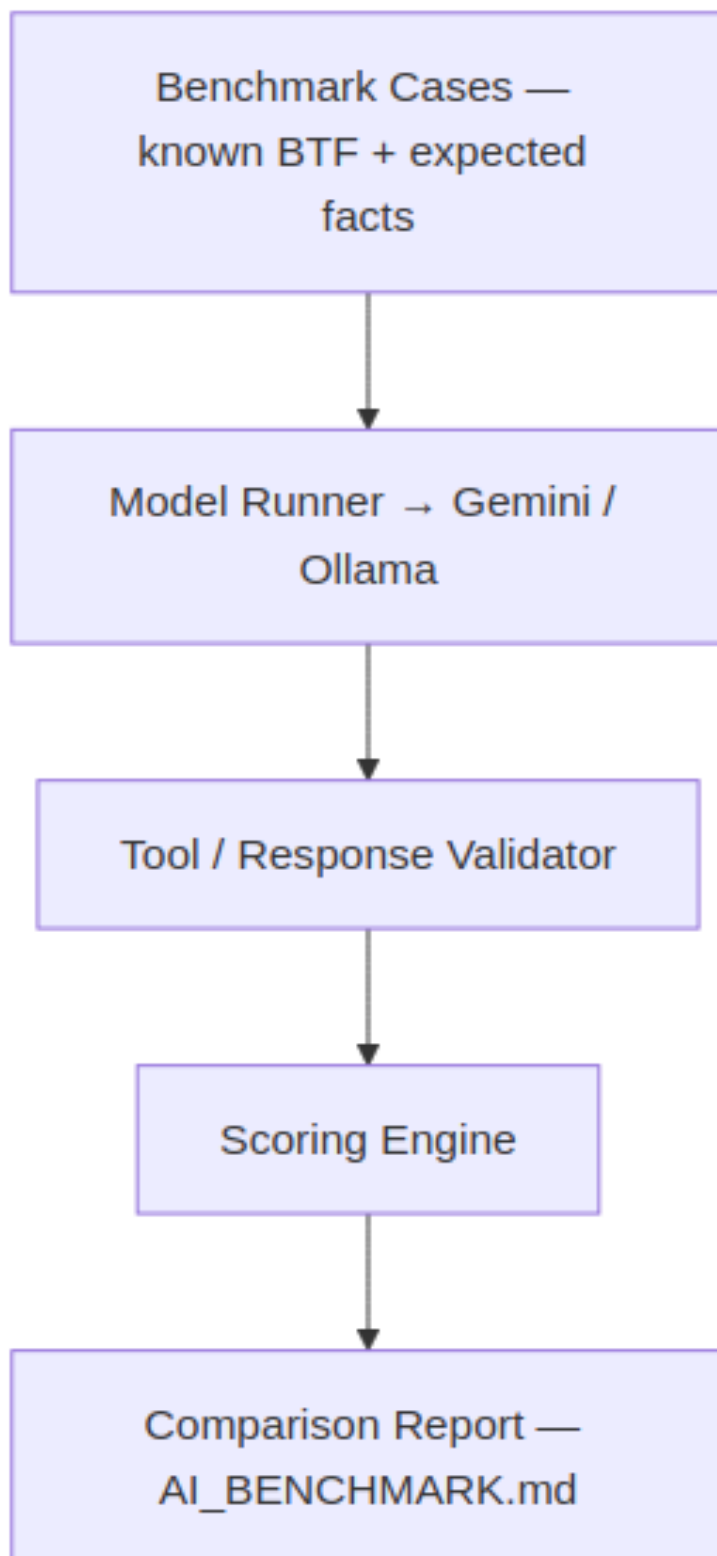
Gemini 3.7 Flash / Gemini 3.5 Flash-Lite
vs
Qwen3.5 9B (shipped default)
Qwen3.8 27B

Does extra local capacity improve investigation quality enough to justify memory and latency?

Reproducibility and architecture

Each run should save timestamp, app version, dataset version, model ids, endpoint config, cases, prompts, tool calls/results, final responses, scores, and timing. A run ID (AI Benchmark #2026-08-13-001) keeps results comparable when model behavior drifts without a viewer code change.

--fail-under N can fail CI when a model drops below a threshold (live often wants 0 so HTTP errors still write the report).



Investigation Case

Desktop and Web share one **Investigation Case** model (btf-investigation-case): question, scope (trace / C1-Cn / tasks / cores), hypotheses with status (**supported** / **possible** / **need evidence** / **rejected**), evidence graph, coverage, falsification checks, conclusion, and validation. Lockstep notes: **Implementation notes**.

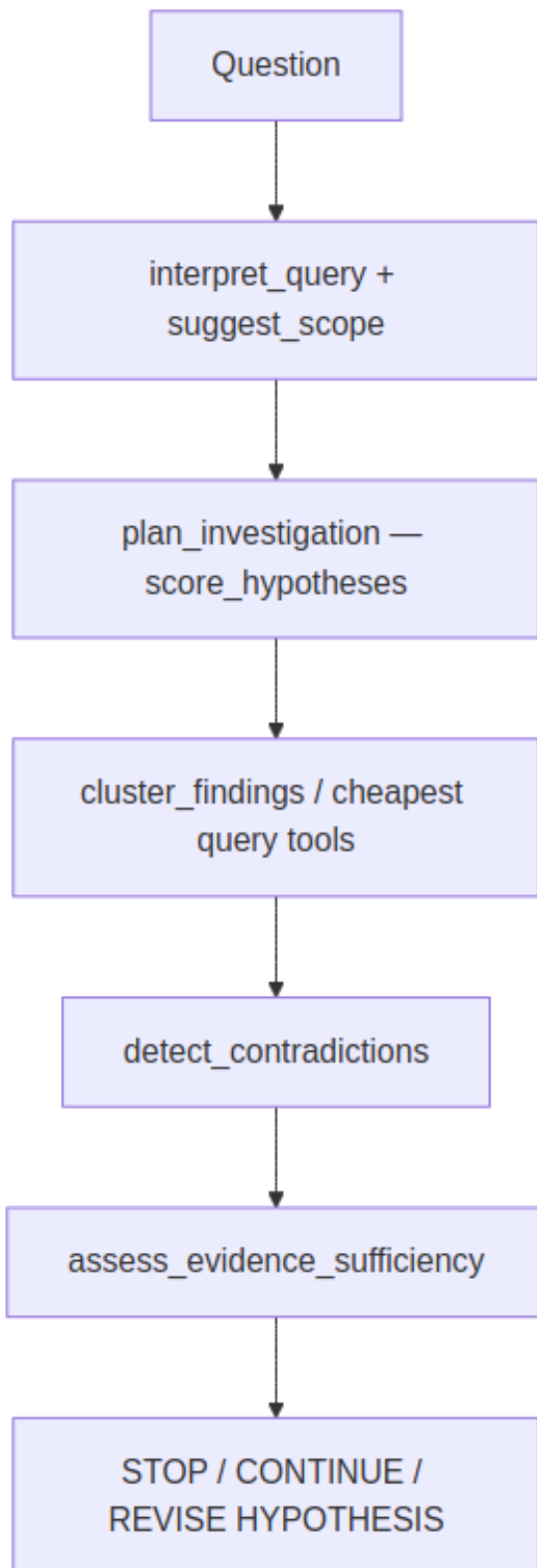
After each final assistant reply a host-side **validator** extracts jump:TIME and Task[id] claims and flags invented names or timestamps outside the cursor window. **Test connection** appends a **model capability** card (live chat / structured output / tool calling overlay on a 3B vs 7B+ heuristic). Headless eval:

```
make -C BTFViewer ai-test  
# or: python builds/btf_viewer.py ai-test --dataset tests/ai --fail-under 70
```

Modes (Quick / Diagnose / Compare / Optimize / Report) map onto existing templates; they do not add new tools. Always confirm on the timeline.

Investigation planner

Host-side planner. **Cheapest evidence first**. User-facing loop: **README → Investigation planner**.

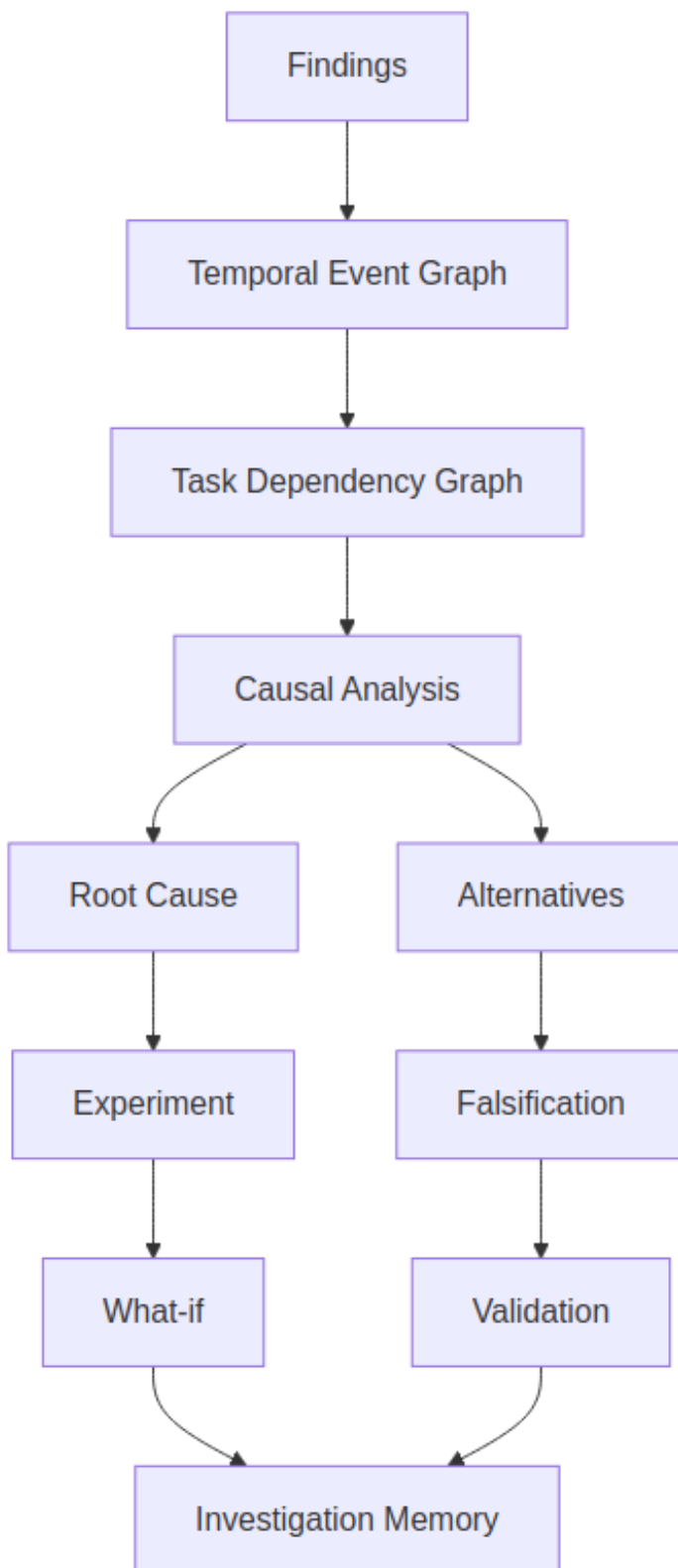


Tool / helper	Host behaviour
<code>plan_investigation</code>	Rank hypotheses and a cheap tool sequence from findings + question
<code>suggest_scope</code>	Task, related tasks, evidence times (or current cursors)
<code>detect_contradictions</code>	SUPPORTED / CONTRADICTED / INSUFFICIENT (e.g. execution $\gg$ blocking vs mutex hypothesis)
<code>assess_evidence_sufficiency</code>	Coverage heuristic $\rightarrow$ stop / continue / revise
<code>score_hypotheses</code>	Evidence-weighted scores (not a GUI tool)
<code>cluster_findings</code>	Group by shared task or pattern
<code>generate_fingerprint</code>	HIGH / MEDIUM / LOW scheduling, sync, timing bands
<code>find_similar_investigations</code>	Jaccard-style match vs recorded experiment outcomes
<code>regression_localize</code>	A vs B deltas $\rightarrow$ task / region / likely mechanism
<code>build_causal_chain</code>	Edges tagged causal / correlated / temporal; disclaimer required
<code>generate_experiment_plan</code>	Ranked pin / contention / priority experiments
<code>record_experiment_outcome</code>	Persist outcome (Desktop [ai] experiment_outcomes, Web localStorage)
<code>score_investigation</code>	Phase 3 extras: evidence_efficiency, investigation_cost, false_confidence, falsification_quality, scope_accuracy, stop_efficiency (also spread into score_benchmark_case, with adversarial rates)

Do **not** add chat templates after `auto_investigate`.

Causal and temporal engines

Host-side heuristics over Analysis Findings — not an RTOS scheduler replay. User-facing loop stays on **Investigation planner**. Diagnose / Investigate / Auto investigate walk explanation tools before experiments: graph → temporal → rank_root_causes → challenge_conclusion, then what_if.



Tool / helper	Host behaviour
<code>analyze_temporal_causality</code>	Happens-before chain from finding times (jump:TIME)
<code>build_task_dependency_graph</code>	BTF sync/preempt/migrate/PI graph (finding-wording fallback); optional task neighborhood
<code>decompose_response_time</code>	Relative delay shares (mutex, preemption, migration, execution, scheduler)
<code>rank_root_causes</code>	Rank hypotheses or finding buckets
<code>verify_claim</code>	SUPPORTED / PARTIAL / UNSUPPORTED vs findings and cursors
<code>challenge_conclusion</code>	Alternatives and missing evidence
<code>investigation_memory</code>	Store/recall (Desktop [ai] investigation_memory, Web localStorage)
<code>cluster_incidents</code>	Group findings by time proximity
<code>close_investigation</code>	Record conclusion and close the case
<code>analyze_distribution</code>	p50 / p90 / p95 / p99 / p99.9, stddev, CV, 3-sigma outlier rate; BTF execution/blocking/PI/tick harvest
<code>analyze_periodicity</code>	Period/jitter from tick, STI, ISR, timer, or task-release times; kind = drift vs jitter vs WCET vs scheduler
<code>summarize_investigation_context</code>	Compact findings, hypotheses, and tools run

Engine limits

Engine	What it is	What it is not
analyze_temporal_causality	Happens-before from finding jump:TIME	Kernel event replay
build_task_dependency_graph	BTF sync / preempt / migrate / PI edges; 2-hop task neighborhood	Full ISR / object graph
decompose_response_time	Relative shares from finding magnitudes	Cycle-accurate milliseconds
rank_root_causes	Hypothesis or finding-bucket rank	A probability
investigation_memory	Local store / recall notepad	Team knowledge base
cluster_incidents	Time-proximity groups	Shared-mutex / causal clustering
close_investigation	Case status closed plus conclusion	Full firmware A/B lifecycle
analyze_distribution	BTF execution / blocking / PI / tick samples (cap 8000)	A response-time series the parser does not have
analyze_periodicity	Inter-arrival jitter and kind	A kernel period timer
simulate_schedule	LEVEL 1 helper inside what_if	A GUI tool or RTOS kernel

Out of scope (do **not** add chat templates for these): trace-to-code (ELF / DWARF), real scheduler or hardware-aware simulation, model routing, automatic benchmark-case generation, natural-language → metric compiler, anomaly discovery without Analysis Findings, shared team investigation database.

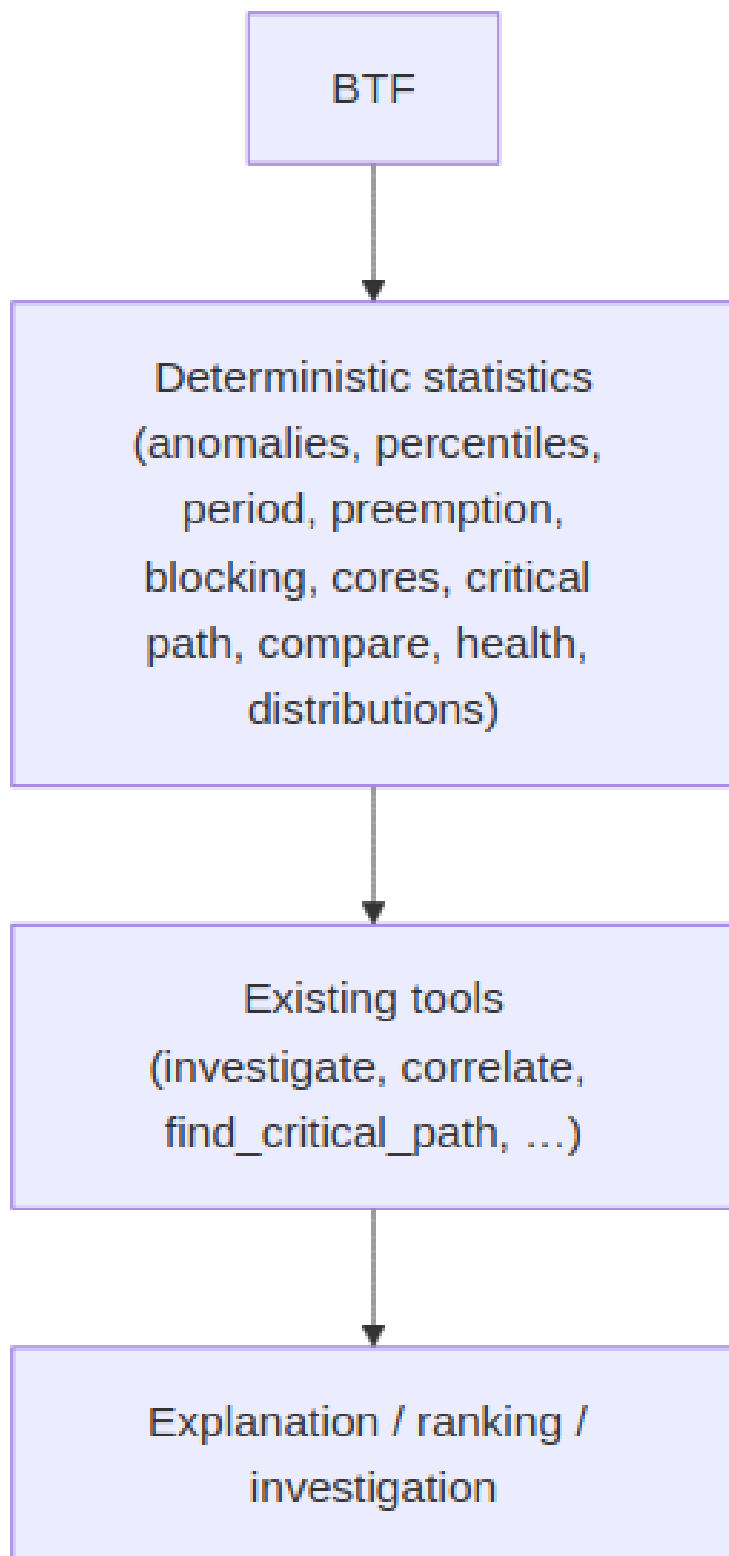
Do **not** add chat templates after auto_investigate. Next gains come from deeper engines, not more buttons.

Implementation notes

Technical notes for keeping Desktop and Web in lockstep. For user-facing Case and Evidence behavior, see [README → Investigation Case](#). For live-suite XML, see [Benchmark and evaluation suite](#). Recorded scores are in [AI_BENCHMARK.md](#).

Analysis vs AI tools

Facts come from BTF Statistics pages first. AI ranks, explains, and navigates those facts. **Do not add AI tools** for work those pages already do (no `detect_timeline_anomalies`, extra jitter tools, or histogram tools). **Do not** invent kernel response time, inspect ELF/source, or simulate the scheduler.



The shipped loop stays **Triage → Investigate → Verify → Correlate → Critical Path → Dependency Graph → Temporal Causality → Rank → Challenge → What-if → Report**. Improve access to Statistics evidence; do not grow the tool list. User-facing page map: [README](#) → [BTF analysis pages](#).

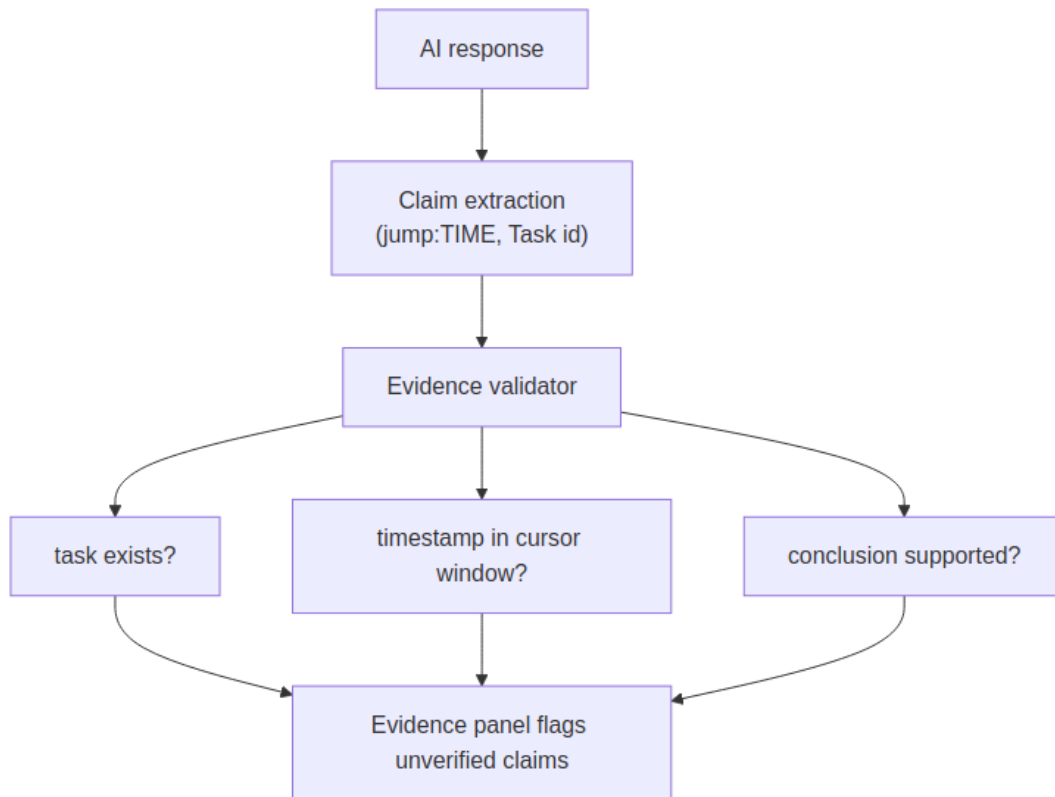
Shared Case / Evidence engines

Desktop and Web share one Case, Evidence, planner, causal, tool, and mermaid implementation. After AI UI changes: make -C BTFViewer bundle and make -C BTFViewer web.

UI lockstep: mode chips wrap. Primary templates are two rows: **Analysis Findings / Explain region / Investigate**, then **Auto investigate + More templates....** Chip min-height 28px, disabled chips/menu items #8a96a8. Findings **Investigate...** uses the same outline style as the other Analysis footer buttons (not accent/primary). **More** templates use the same groups in a 2-column overlay. Findings **Save recipe...** and **Story...** stay on that dialog. Trace Compare opens from toolbar **Compare**, not the Statistics footer.

The Desktop ai-test CLI and the Web offline benchmark share tests/ai fixtures (tracked .btf stubs + dataset.json). Live runs accept --context-mode and --compare-context (see [Context mode benchmarking](#)).

Validator



The host validator runs after the final reply. Prompting still forbids inventing numbers, task names, and `jump:TIME`; the validator is the guard, not the prompt.

Experiment close-out

`validate_experiment` compares expected vs actual signed percents (VALIDATED / PARTIALLY VALIDATED / DISPROVED), updates open hypotheses, and offers **Save to knowledge** (`btfexp:save`). Empty actual is filled from the last Trace Compare refresh (including **Scope to cursors**) or `compare_performance` via `experiment_percents_from_compare`. Toolbar **Compare** → **Validate experiment...** closes the dialog and asks the model to call the tool with actuals omitted. Firmware-change capture remains a user step (new trace + toolbar **Compare**).

Capability, cost, privacy, knowledge

Feature	Host behaviour
Capability probe	Test connection lists models, chats with a JSON structured-output probe, then tool-calling (btf_ping then btf_pong). Live results overlay chat / structured output / tool calling / multi-tool chaining; long context and reasoning stay heuristic.
Cost	A dedicated usage bar shows Context: Compact · 4.6k tok · 3 tools · 12s (mode, tokens, tools, model time). Evidence uses the full format_cost_meter line. Clear resets replies, the meter, and current investigation issues.
Privacy	Chip • Local / • Cloud / • Sensitive. Cloud send is blocked when sensitive; otherwise annotations are sanitized and optional task-name aliases apply (apply_cloud_privacy).
Knowledge	investigate matches user-saved entries (More → Save current finding...), then baseline, then the builtin catalog. Typical vs current rates show when both exist.
Interpret	Free-form Ask host-interprets (interpret_query), shows the scope card, then auto-runs (same as Run investigation). Templates / modes / prior assistant replies / short follow-ups skip the host interpret step. Scope toggles still allow re-run.
Tool Why?	Evidence Investigation lists each tool with a host-side reason (btf_tool:why/name).

Diagrams

Replies may include Mermaid sequence diagrams for mutex, blocking, and priority events, or flowcharts for core migrations. **Compact** context mode emits diagrams only when the user asks. Pipe **Markdown tables** and sanitized HTML tables from Findings render as tables in the reply pane. In-chat Markdown / HTML tables follow the current theme. The Evidence panel also generates an Investigation tree when investigate returns a root-cause chain. Diagrams follow the current light or dark theme; **Save As...** HTML exports use the light palette.

- Click a **task** node to lock-highlight that timeline row (Low[266] (Core 0) resolves to Low[266]).
 - Click a **core** node (Core_0, C0, C1) to switch to Core View and scroll to that core.
 - Mutex hex and other unresolved labels do nothing (the timeline stays undimmed).
 - Click empty figure area to open a larger zoom window (scroll to zoom 0.5–6×; **Esc** or **Close**). Trackpad pinch is treated as scroll.
 - The link row under the figure lists only `jump:TIME` and resolvable task/core names — not finding titles, hypotheses, or graph node ids (F, C0, ...).
 - **Save As...** HTML keeps inline SVG with clickable nodes (chat zoom wrappers are omitted).
-

Documentation navigation

Document	Question answered
README.md	How do I use BTFViewer?
WORKFLOWS.md	How do I diagnose a problem?
STATISTICS.md	What does this measurement mean?
AI.md	How does AI-assisted investigation work?
